

Core-Pitch Conversion Technique for Randomly Coupled Multi-Core Fibre Links by Cladding Mode Coupling

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Abstract We propose a core-pitch conversion technique using only a simple spliced-taper structure of randomly coupled multi-core fibres (RC-MCF). We demonstrate up to 20- μm core-pitch conversion with loss below 0.9 dB and low mode-dependent loss, which can connect RC-MCF transmission line and smaller-core-pitch amplifier. ©2026 The Author(s)

Introduction

To accommodate the growth in data traffic, space-division multiplexing (SDM) technologies that increase the transmission capacity per fibre have been actively studied. Randomly coupled multi-core fibres (RC-MCFs) are particularly promising as high-density SDM fibre because they have much higher core density than weakly coupled MCFs (WC-MCFs) to allow mode coupling by utilizing multiple-input multiple-output (MIMO) processing [1]. RC-MCF exhibits low spatial mode dispersion (SMD) characteristics when the core pitch is approximately 16-25 μm , which reduce the complexity of multiple-input multiple-output (MIMO) processing [2].

To establish optically repeated transmission systems using RC-MCFs, RC-multi-core erbium-doped fibre amplifiers (MC-EDFAs) are advantageous for reducing power consumption with cladding pumping scheme [3]. MC-EDFAs are more efficient than parallel arrays of single-core EDFAs when the core pitches of around 15 μm or less, which is narrower than the design range of RC-MCF transmission line [3]. Therefore, core pitch conversion technique is required.

Some core pitch conversion techniques have been demonstrated by utilizing gradient-index (GRIN) lenses [4] or bridge fibres with a reduced mode-field diameter (MFD) [5]. However, these methods require the optimisation of the lens or bridge-fibre design depending on the core pitch of the fibres used.

In this work, we propose a core-pitch conversion technique by converting the core modes into cladding modes using a tapered structure, which requires a simple spliced-taper structure of multi-core fibres with different core pitch. Our simulations and experimental results show its applicability to a wide range of core pitch difference with almost the same taper structure, with the insertion loss of less than 0.9 dB and sufficiently low mode-dependent loss.

Simulation

Figure 1 (a) shows the diagram of the proposed core pitch converter. The two-core RC-MCFs with

different core pitches of Λ_1 and Λ_2 are tapered after rotationally aligned splicing. The taper profile was assumed that has a diameter varying quadratically along the fibre longitudinal direction [6]. Here, the main design parameters of the taper profile are the taper length L and ratio R , where L is the length of the section in which the cladding diameter changes from the original cladding diameter D_{clad} to the minimum value D_{min} , and R is the ratio of D_{min} to D_{clad} . In Fig. 1, parameters were set as $L = 10$ mm, $R = 0.3$, $D_{clad} = 125$ μm , $\Lambda_1 = 15$ μm and $\Lambda_2 = 40$ μm . Figure 1 (b) shows the simulated electric field distribution. Unless otherwise noted, the following simulations assume a two-core fibre with a core radius of 4.2 μm , a cladding diameter of 125 μm , a wavelength of 1.55 μm , and a relative index difference Δ of 0.35%. Light is launched into one core, which is equivalent to equally launching the even and odd supermodes. As shown in Fig. 1 (b), a down-taper structure converts an input core mode into a cladding mode, then an up-taper structure reconverts it into a core mode at the end. This result indicates the core pitch can be converted via cladding modes by the proposed tapered structure. Note that this method is effective for RC-MCF because cladding mode conversion induces strong inter-core-mode coupling.

Figure 2 shows the simulated coupling loss of

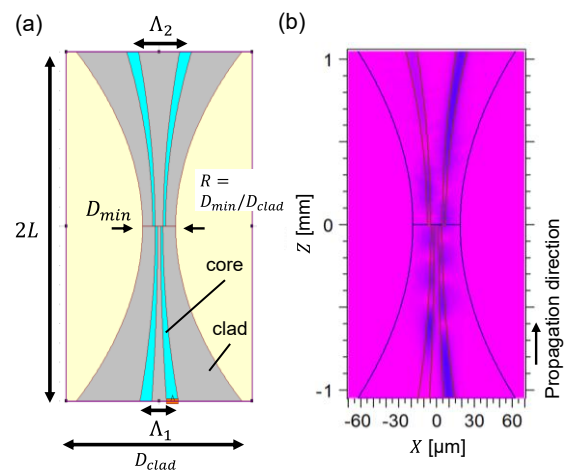


Fig. 1: (a) Diagram of the proposed method and (b) simulated electric-field distribution.

the converter as a function of R and L with $\Lambda_1 = 15 \mu\text{m}$ and $\Lambda_2 = 25 \mu\text{m}$, which correspond to the minimum and maximum core pitches reported for RC-MCFs, respectively [1]. Figure 2 shows that the loss tends to decrease as the R becomes smaller. This is because cladding-mode coupling is enhanced as $D_{\min}$ decreases, reducing the splice loss caused by the core position mismatch (nearly 4 dB in Fig. 2). The loss increases in $R < 0.2$ may be caused by the light leaking due to excessively reduced cladding diameter [7]. The loss also seen that the loss tends to decrease as L becomes longer. This suggests that the adiabatic variation is effective for achieving low loss, which is similar to in conventional taper fabrication. Within the range shown in Fig. 2, the loss was minimized at $R = 0.3$ and $L = 11 \text{ mm}$, which was defined as the minimum-loss condition in this paper.

Next, the loss dependence on the core-pitch difference $\Delta\Lambda (\equiv \Lambda_2 - \Lambda_1)$ was evaluated. Figure 3 (a) shows the loss as a function of $\Delta\Lambda$. Here, Λ_1 was fixed at 15 and $\Lambda_2 = \Lambda_1 + \Delta\Lambda \mu\text{m}$ was varied from 20 to 40 μm . The loss value in Fig. 3 (a) was the loss value at the minimum-loss condition defined above. The taper shape (R and L) at the minimal (optimised) loss point is shown in Fig. 3 (b). Figure 3 (a) shows that the loss weakly depends on $\Delta\Lambda$ and it remains below 0.2 dB in all cases, indicating the effect of cladding mode coupling. If light propagation were governed by core modes, simple splicing of multi-core fibres with different core pitches would result in lateral misalignment of each core. As a result, the loss increases quadratically with $\Delta\Lambda$ [8] as described by Marcuse equation. As shown in Fig. 3 (b), the optimum values of L and R weakly depends on the $\Delta\Lambda$, suggesting that nearly the same taper shape can be used over a wide range of $\Delta\Lambda$.

Experiment

Based on the above numerical investigation, the proposed core-pitch conversion technique using cladding-mode coupling was experimentally verified. Table 1 shows the parameters of the MCFs used in this experiment. All have two pure-silica step-index cores with the different core pitches but the same cladding diameter of 135 μm . The experiments were carried out for all combinations of the three MCFs to investigate the properties over a $\Delta\Lambda$ range sufficiently larger than the maximum $\Delta\Lambda$ among RC-MCFs.

The experimental setup is shown in Fig. 4. In the experimental system, the loss and the root mean square mode-dependent loss (MDL_{rms}) of the converter were measured. For the loss measurement, light from a superluminescent diode (SLD) was launched into one core of a RC-MCF. Power fluctuation due to optical interference was

suppressed by the broad spectrum of the SLD (3 dB bandwidth: 1490-1560 nm) and sufficiently long (more than 500 m) RC-MCF (MCF-A). The loss of the converter was evaluated by the cut-back method. For the MDL_{rms} measurement, the light source was switched to a tunable laser source (TLS), and a scrambling method [10] was carried out. It should be noted that the loss and MDL_{rms} were measured for the same sample. The inset of Fig. 4 shows the tapering procedure. The MCFs were rotationally aligned so that their core arrangements were similar, and then fusion-spliced. After that, heating was applied around the splice point, and a tapered shape was formed by melt-stretching with a heater. By the tapering machine used here, quadratic-like taper shape

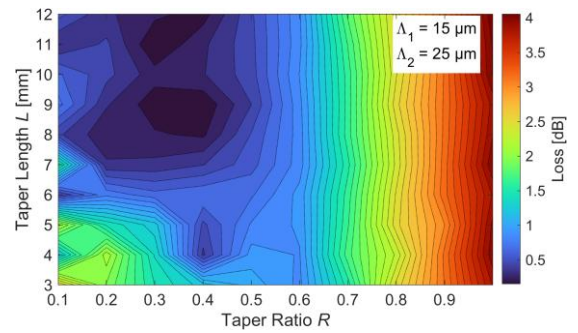


Fig. 2: Simulated loss as a function of taper shape (taper length L and taper ratio R).

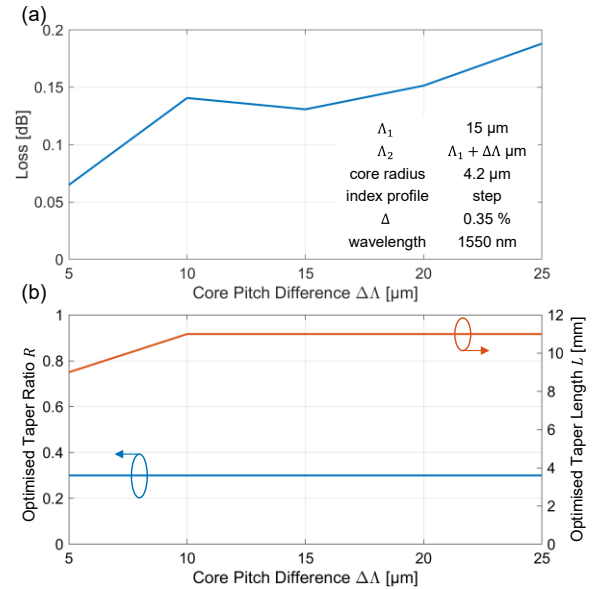


Fig. 3: (a) Simulated loss as a function of core pitch difference. The loss was minimized by L and R . (b) Simulated optimised taper ratio R and taper length L .

Tab. 1: Parameters of the two-core fibres under the tests.

Number	Core Pitch [μm]	Core Radius [μm]	Δ [%]
#1	19.8	4.5	0.38
#2	25.2	4.4	0.35
#3	39.4	4.3	0.36

was fabricated.

Figure 5 shows the measured shape of the fabricated taper of the #1-#2 samples as an example. The outer-diameter profile was extracted from images acquired with a microscope. As shown in Fig. 5, the obtained shape had no significant error and could be well approximated by a quadratic function.

Figure 6 shows the measured loss of the fabricated taper, together with the calculated minimal loss with the parameters of the actual fibres used in the range of Fig. 2. The points represent the mean values obtained from three fabricated samples ($N = 3$). The upper and lower bounds of the error bars denote the maximum and minimum loss values of the fabricated samples, respectively. As shown in Fig. 6, the calculated and experimental losses were consistent within the same order of magnitude and did not increase quadratically along the $\Delta\lambda$ (as discussed in Fig. 3 (a)). These results may indicate that pitch conversion was successfully achieved via cladding-mode coupling. The MDL_{rms} was also measured and the MDL_{rms} was below 0.1 dB at all cases and samples (not plotted).

The measured loss was higher than the calculated value and varied among the samples. This discrepancy may be due to shape errors during stretching caused by misalignment between the splice point and the heating centre. This caused a mismatch between the splice point and the position of the minimum cladding diameter of the taper, effectively changing the taper shape. From the calculation (not shown), the variation of approximately 0.2 in R increases the loss by about 0.2-0.3 dB. In this estimate, the effect of variations in L was neglected because misalignment causes the effective L to increase on one side of the splice and decrease on the other, so the net effect is expected to be small. Therefore, further loss reduction may be achieved using an integrated apparatus for fusion splicing and taper fabrication.

The mean loss values in each $\Delta\lambda$ case were 0.9 dB or less, which are comparable with or lower than those in previous pitch conversion techniques [4, 5]. In terms of MDL, measured MDL_{rms} was sufficiently smaller than the reported MDL per span in transmission experiments [11], which suggests the proposed technique is applicable to the connection between amplifiers and transmission fibres.

These results demonstrate low-loss, low-MDL core-pitch conversion between RC-MCFs using cladding-mode coupling. In particular, case #3-#1 shows that the proposed technique enables RC-MCF/WC-MCF interconnection, allowing mature WC-MCF optical components to be used in RC-

MCF systems (e.g. low-loss bundle-type WC-MCF fan-in/fan-out (FIFO) devices [5] can be used for RC-MCF link via proposed spliced-taper technique).

Conclusion

We have shown, through both simulations and experiments, that MCFs with different core pitches can be connected with low loss by inducing cladding-mode coupling in a tapered region. These results showed a loss of 0.9 dB or less and an MDL_{rms} of 0.1 dB or less for a pitch difference of up to 19.6 μm with the proposed a core-pitch conversion method. This method will be effective for connecting RC-MCFs and MC-EDFAs with different core pitches, enabling us to construct a transmission systems with both low-SMD and energy-efficient. Wide-range pitch conversion also enables adaptation of matured optical components for WC-MCF to RC-MCF systems.

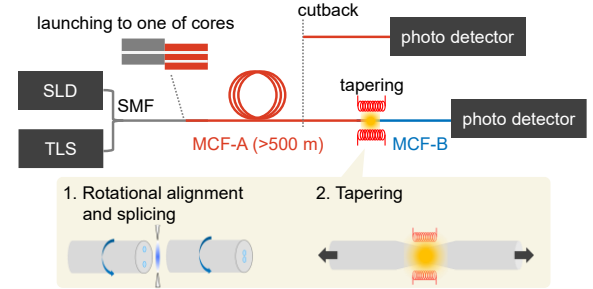


Fig. 4: Experimental Setup. Inset shows tapering procedure.

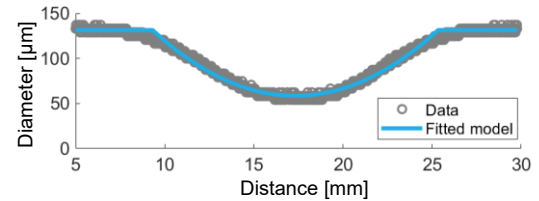


Fig. 5: Measured taper shape (Case of #1-#2 connection)

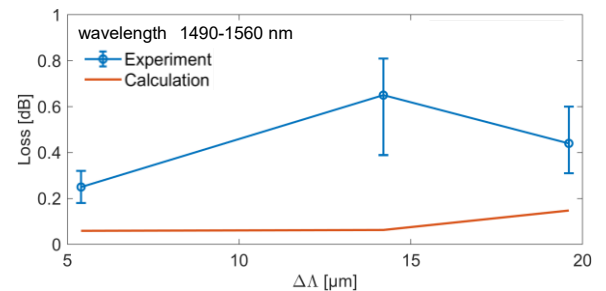


Fig. 6: Measured loss values of the fabricated samples and calculated loss values

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